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2. The table shows some physical properties of Group 6 elements.

Element	Melting point (°C)	Boiling point (°C)	Density (g/cm^3)	Electrical conductor
oxygen	–219	–183	0.0014	no
sulfur	115	445	2.0	no
selenium	221	685	4.8	semi-conductor
tellurium	450	988	6.2	semi-conductor

(a) (i) Describe the trend in the melting points of the Group 6 elements. [1]

....

(ii) Give the physical state of selenium at 400 °C. Give a reason for your choice. [2]

....

....

(iii) Explain why it is difficult to classify selenium as either a metal or a non-metal. [1]

....

....

....

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(b) Selenium dioxide reacts with sodium hydroxide to produce sodium selenite and water.

$$\underset{\text{selenium dioxide}}{SeO_2} + 2NaOH \longrightarrow \underset{\text{sodium selenite}}{Na_2SeO_3} + H_2O$$

(i) Calculate the relative formula mass (M_r) of sodium selenite, Na_2SeO_3. [1]

$A_r(Na) = 23$ $A_r(Se) = 79$ $A_r(O) = 16$

M_r = ...

(ii) Calculate the percentage by mass of selenium in sodium selenite. [2]

Percentage = ... %

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